

SQL Lecture Notes (5.19.2026)

1. SQL Architecture:

- i) **SQL Server:** software that stores, manages and processes data.
- ii) **Workbench:** A graphical tool used to interact with and manage databases.
- iii) **Database:** An organized collection of data stored electronically.

2. Schema:

The blueprint of a database that defines tables, columns, relationships and data types.

3. SQL Command Types

- i) **DDL - Data Definition Language:** Used to define or modify the structure of the database. (CREATE, ALTER, DROP, TRUNCATE)
- ii) **DML - Data Manipulation Language:** Used to manipulate data inside tables. (INSERT, UPDATE, DELETE)
- iii) **DQL - Data Query Language:** Used to retrieve/query data from tables. (SELECT)
- iv) **DCL - Data Control Language:** Used to control access and permissions to database objects.
- v) **TCL - Transaction Control Language:** Used to manage transactions and maintain data integrity.

4. Constraints:

Rules that restrict invalid data in a table.

- **NOT NULL Constraint:** Ensures that a column cannot store empty or null values.
- **UNIQUE Constraint:** Ensures that all values in a column are different from each other.
- **Primary Key (PK):** Uniquely identifies each record in a table and cannot contain null values.
- **Foreign Key (FK):** Creates a link between two tables using a common column.

- **CHECK Constraint:** Restricts values in a column based on a specified condition.
- **DEFAULT Constraint:** Automatically assigns a predefined value when no value is provided.

5. Primary Key and Foreign Key:

i) **Primary Key:** A column that uniquely identifies each row in a table. It is also known as parent or referenced table.

ii) **Foreign Key:** A column that connects one table to the primary key of another table, creating a relationship between them. It is also known as child or referencing table.

6. Metadata:

Data that provides information about other data. It describes details like table names, column types, relationships and database structure.

7. Drop, Delete and Truncate:

i) **Drop:** permanently removes a table and its structure from the database.

ii) **Delete:** Removes selected rows from a table and can be rolled back.

iii) **Truncate:** Removes all rows from a table quickly while keeping the table structure intact.